/ Heuristics in Chosing Activation Functions:

A. use $y = \tanh(x)$ in hidden layer,

if we use $\tanh(x)$ as activation functions for all perceptrons in single layer, the outputs of them will all center around 0 ($\because$ threshold of $y=\tanh(x)$: $y=0$),

we use these set of outputs as the set of inputs for one of perceptrons in next layer, the value of z for this perceptron will centered around 0,

z value around 0 has larger derivative of z and is easier to compute gradient descent in neural network.

B. use $y = S(x)$ in output layer, because the y range $0 \sim +1$ of $y = S(x)$ can be interpreted as a probability.